

Irene García Muñoz

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Profile

I am a PhD student based in the Manchester Department of Mathematics. My research to date has used mathematical modelling to address questions in public health, with a particular focus on modelling infectious diseases.

Current Position

2023–present	PhD, Department of Mathematics, University of Manchester, UK Thesis title: <i>Respiratory Viruses in Care Homes: Development of Transmission.</i> Supervisors: Thomas House, Ian Hall (University of Manchester) & Maria Zambon (UKHSA).
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Education

2019–2023	Master of Mathematics (MMath), Department of Mathematics University of York, UK Group research project: Calculus of variations, Noether's theorem and Classical Mechanics. Supervisor: Prof M. Pushkin. Individual research project: Mathematical and Bacterial Billiards. Supervisor: Prof K. Ilin.
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Teaching

2023– present	Teaching Assistant, University of Manchester Responsible for planning and leading weekly problem sessions for foundation and first-year undergraduates from different STEM backgrounds. Modules included Analysis, Calculus, Classical Mechanics, and Statistics.
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Skills and Other Training

Mathematical Modelling	Epidemic modelling, Markov processes, stochastic processes, statistical inference.
Programming	Python, R, Matlab.
APTS	Academy for PhD Training in Statistics.
Representative	Elected Mathematics Department representative third time in a row at the University of York (2020-current), to represent over 1400 students and lead the team of Course Representatives to make sure that the student's voice is listened to and acted upon by the department.

Other Professional Activities

June-July 2022	Cardinal Hume Catholic School, Gateshead Teaching Assistant in the Department of Mathematics. Assisted in the delivery of instructions to students as a whole class, in groups and individually. Planned and facilitated workshops for students. Attended staff development training sessions.
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Research Impact

2024–Present	Contributions to the UK's Management of Care Home Hazards Outbreak Working Group (MOCHHO), advising on outbreaks in care homes.
2024–2025	Contributions to the WHO Somalia smallpox project, advising on mathematical modelling approaches.

Awards and grants

June 2026	Doctoral Training and Development Support Fund. Awarded by the FSE PGR Panel to facilitate conference attendance or support further training and development. Value: £500.
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Presentations and Conference Contributions

March 2026	Contributed talk: Inference for Within- and Between-Partnership Transmission Rates for HIV Infection. British Applied Mathematics Conference (BAMC), Norwich, UK.
March 2026	Contributed talk: Progress update on PhD studies on developing transmission models of respiratory viruses in Care Homes. Adult Social Care Symposium (ASC Symposium), London UK.
July 2025	Contributed talk: Estimating HIV Transmission Dynamics in Serodiscordant Couples Using a Stochastic SI Pair Model. Young Statistician Meeting (YSM), Liverpool, UK.
February 2025	Contributed talk: Progress update on PhD studies on developing transmission models of respiratory viruses in Care Homes. Adult Social Care Symposium (ASC Symposium), London UK.
September 2024	Contributed talk: Respiratory Viruses: Modelling and Transmission. Poster: Respiratory Viruses: Modelling and Transmission.
July 2024	Mathematics Research Students' Conference (MRSC), Manchester, UK. Contributed talk: Respiratory Viruses in Care Homes: Development of Transmission. European Conference of Mathematical and Theoretical Biology (ECMTB), Toledo, Spain.
September 2023	Poster: Calculus of Variations. Mathematics Research Students' Conference (MRSC), Manchester, UK.

Preprints (Not Yet Peer Reviewed)

June 2026

I. García Muñoz, Joe Flannagan, Emma Louise Fairbanks, Nyall Jamieson, Golsa Sayyar, Jingsi Xu, Maria Zambon, Ian Hall
Transmission Models of Respiratory Infections in Care Homes: A Systematic Review
medRxiv

March 2026

I. García Muñoz, Ian Hall, Thomas House
Inference for Within- and Between-Partnership Transmission Rates for HIV Infection.
arXiv